Intro to Software Engineering

LECTURE : SOFTWARE PROJECT MANAGEMENT

Software Project Management

Software & Project Management

- Corporate America spends more than $275 billion each year on approximately 200,000 application software development projects
- Most of these projects will fail for lack of skilled project management

Management problems were more frequently dominant cause than technical problems

Schedule overruns were more common (89%) than cost overruns (62%)

KPGM's Survey in UK

Software Project Management (2)

Success Factors

1. User involvement – 20 points
2. Executive Support – 15 points
3. Clear Business Objectives – 15 points
4. Experienced Project Manager – 15 points
5. Small milestones – 10 points
6. Firm basic requirements – 5 points
7. Competent staff – 5 points
8. Proper planning – 5 points
9. Ownership – 5 points
10. Others – 5 points

Most of these points are of Management concern

Software Project Management (3)

Primary causes of software runaway

- Project Objectives not fully specified
- Bad planning and estimating
- Technology new to the organization
- Inadequate/No project management methodology
- Insufficient senior staff on the team
- Poor performance by Supplier of hardware/software

In a nutshell

"Organizations that attempt to put software engineering discipline in place before putting project management discipline in place are domed to fail"

Project

"A temporary endeavor undertaken to create a unique *product*, *service* or *result*"

A project is a temporary endeavor undertaken to create a unique product or service

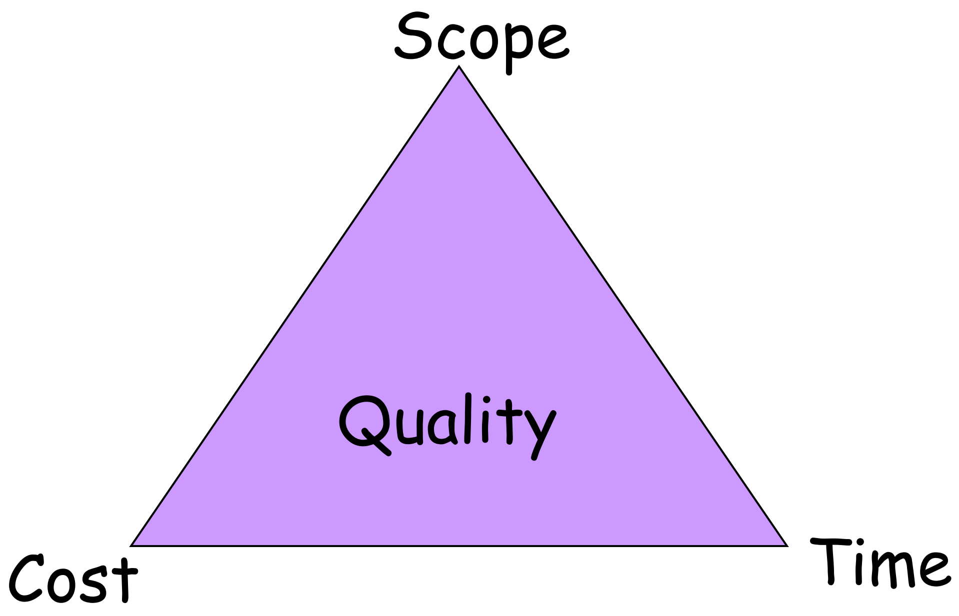
Project Management Institute

Project Management

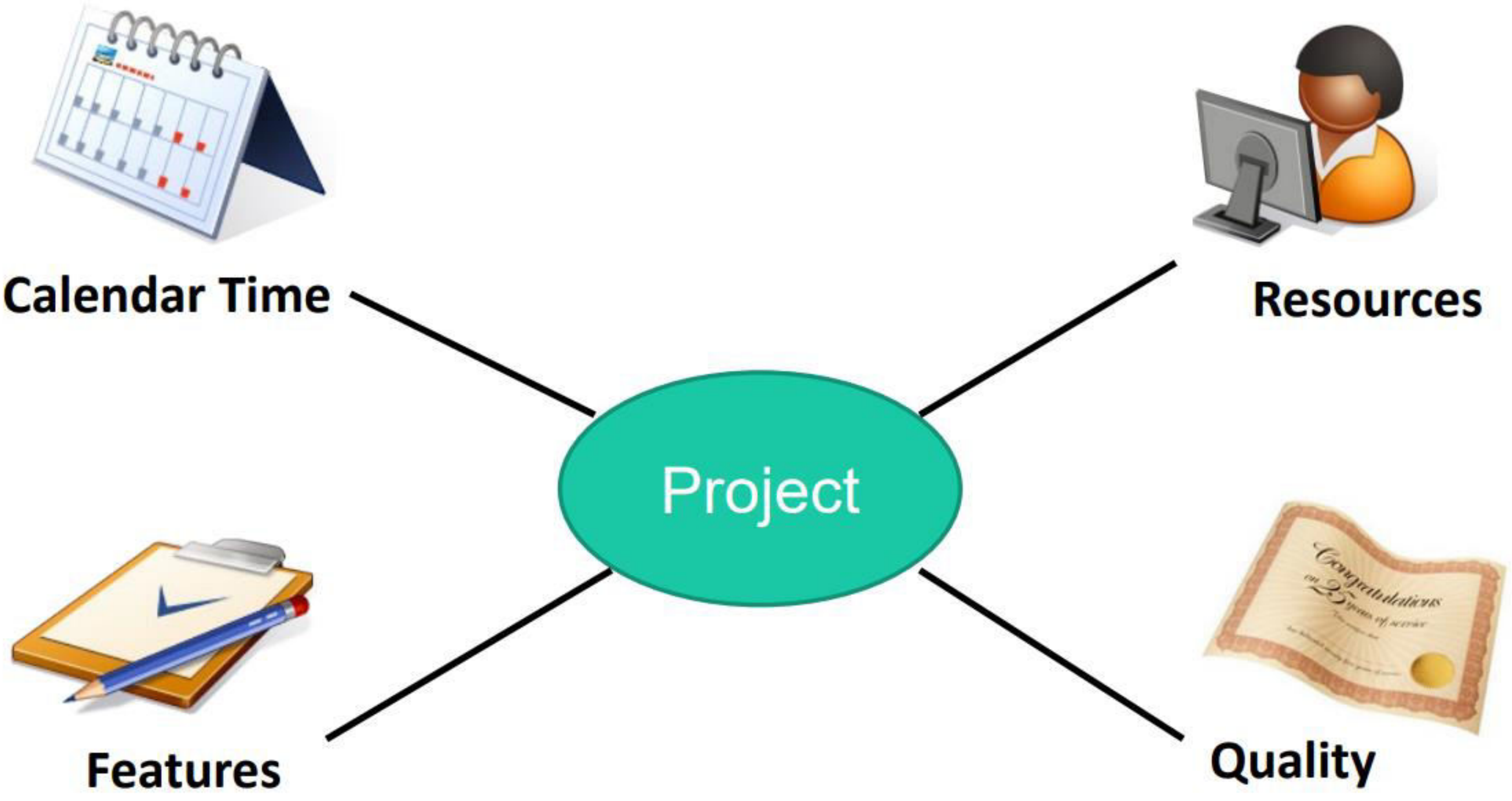
Project management is

The art of managing projects to a successful completion.

The Project Management Triple Constraint:



The four dependent project parameters



Terminology : Deliverable

Work product that is provided to the client

- mock-up,
- demonstration,
- prototype,
- report,
- presentation,
- Documentation
- code
- Release of a system or subsystem to customers or users.

Terminology : Milestone

- Completion of a specified set of activities (e.g., delivery of a deliverable, completion of a process step)
- A milestone is a checkpoint in the project
    - A milestone has
        - A description of the point
        - A date
        - A person or persons who are responsible
- A milestone may or may not have a deliverable associated with it.
    - A deliverable has
        - A description of a deliverable (scope)
        - A date (time)
        - A person or persons who are responsible (cost)

Terminology : Work Package

A **work package** is a large, logically distinct section of work:

– typically at least 12 months duration;

– may include multiple concurrent activities;

– independent of other activities

– but may depend on, or feed into other activities; – typically allocated to a single team.

Terminology : Work Product

A **work product** is a report, diagram, or collection of documents used by the business analyst during the requirements development process.

A **work product** may or may not become a deliverable.

It can be used to share information with stakeholders, elicit requirements, provide status

work product includes the beginning stages of a project, proposals, agendas, reports, analysis and information on the project. A deliverable, on the other hand, is the end result of the project.

Terminology : Task

A **task** is typically a much smaller piece of work:

- A part of a workpackage.
- typically 3–6 person months effort;
- may be dependent on other concurrent activities;
- typically allocated to a single person.

Terminology : Activity

– Must have a clear start and a clear stop

– Must have a duration that can be forecasted

– May require the completion of other activities before it begins

– Should have some deliverables for ease of monitoring

Terminology

Event

The end of a group of activities, e.g., agreement by all parties on the budget and plan

Dependency

An activity that cannot begin until some event is reached

Resource

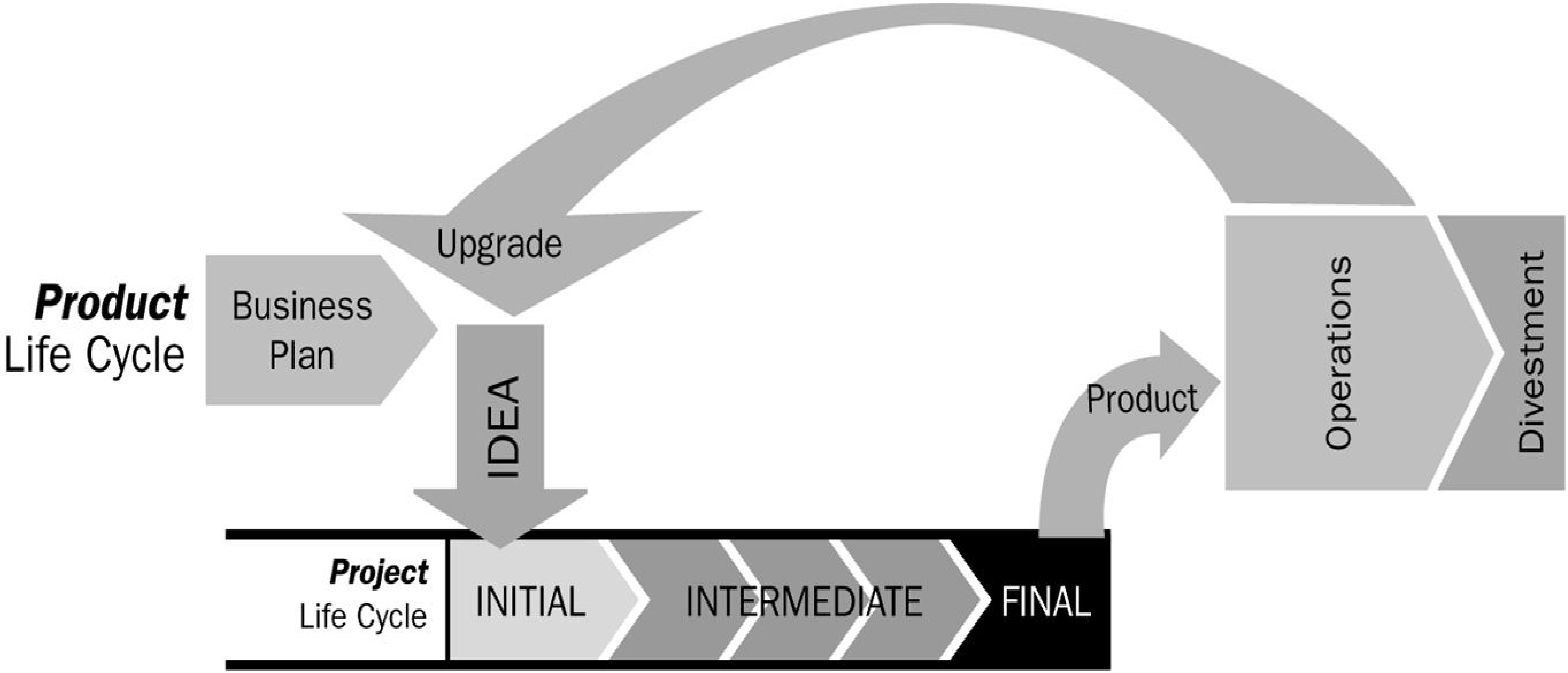
Staff, time, equipment, or other limited resources required by an activity.

Standard Approach to Project Management

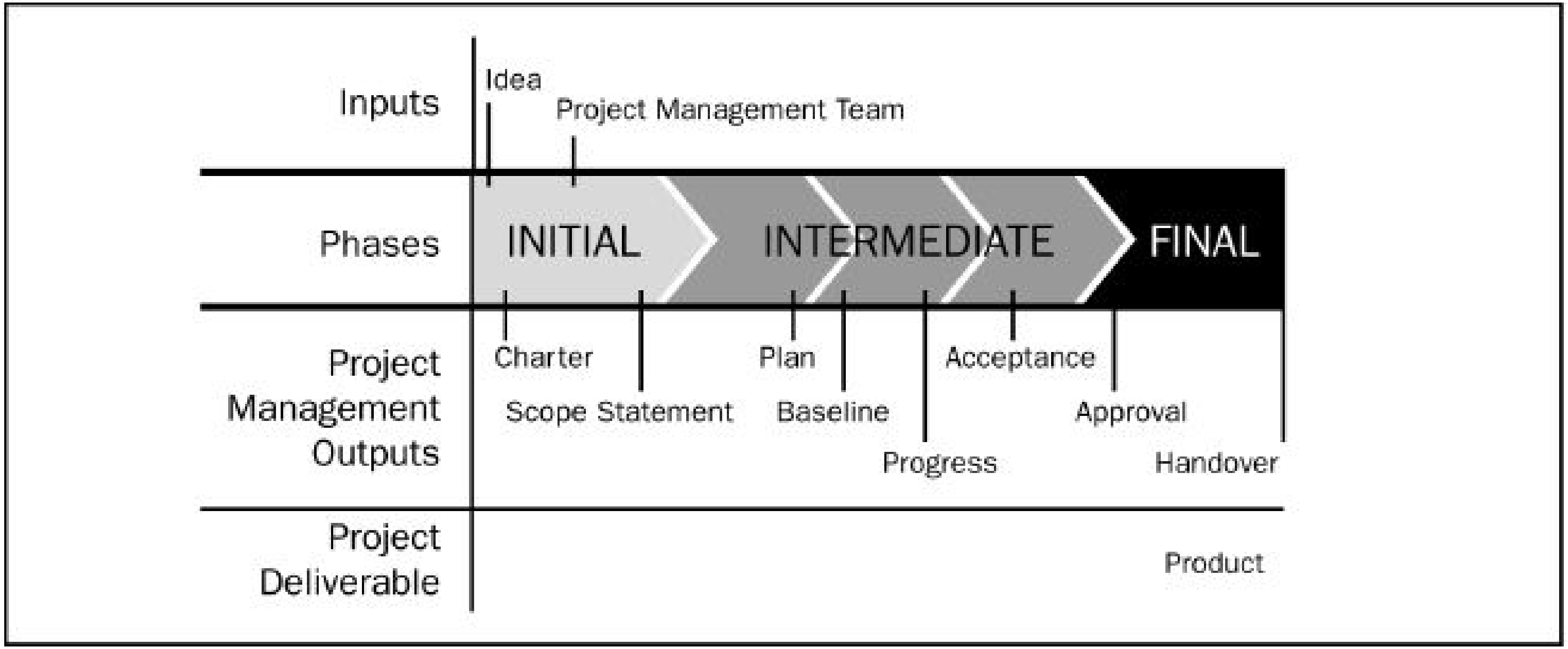
- The scope of the project is defined early in the process.
- The development is divided into tasks and milestones.
- Estimates are made of the time and resources needed for each task.
- The estimates are combined to create a schedule and a plan.
- Progress is continually reviewed against the plan, perhaps weekly.
- The plan is modified by changes to scope, time, resources, etc.

Typically the plan is managed by a separate project management team, not by the software developers.

Product and the Project Life Cycles



Phases in a Project Life Cycle



Project Charter Example: Web Site

One page!

www-cad Project Overview

This project is to create a new web site for the CAD group faculty. The current website at www-cad.eecs has a very old look, it needs an update so that we can attract new students.

Project Approach

The project is a fairly small website based partly on a preexisting site, so we will use a classic waterfall approach with milestones. The project team will consist of the following people. I've estimated the maximum amount of time we can get from each person over the life of the project.

Kurt Keutzer	(2 hrs week for 6 weeks)	Ken Lutz	(2 hrs/week for 6 weeks)
Brad Krebs	(10 hrs/week for 6 weeks)	Christopher Brooks	(10 hrs/week for 6 weeks)
Allen Hopkins	(5 hrs/week for 6 weeks)	Carol Sitea	(1 hr/week for 6 weeks)

The project sponsor is Professor Keutzer. Professor Keutzer is on sabbatical this semester, but we hope to get feedback from him on a continuing basis.

Project Objectives

- Update the look and feel of the website to a modern standard
- Provide access to student and faculty pages
- Provide access to active projects
- Provide access to summaries, downloads and key papers of inactive projects. The old pages of inactive projects should be archived.
- Provide a simple static listing of seminars. A more complex calendar and a search engine are deferred due to schedule constraints.

Major Deliverables

- A schedule along with time estimates.
- A prioritized list of features.
- An example of the main page so we can review look and feel.
- An archive of the old website
- The final website.

Constraints

Professor Keutzer would like to see the web site completed by mid-March: that is when students start looking at graduate schools. Developers might not have much time to work on this project. The project requires timely feedback from the faculty.

Risk and Feasibility

The primary risk is that the project takes too long to complete and we miss the mid-March opportunity. Another risk is that we complete the project too quickly and quality suffers. A third risk is that there are only so many resources available. By fast tracking, we can handle some of the tasks in parallel and avoid these risks. The project is definitely feasible if we roll out the website in stages.

Project Management Life Cycle

Processes/Activities of the **5 phases (Process Groups 2004)** are as:

1. Scope the Project
- State the Problem/ Opportunity
- Establish the project goal
- Define project Objectives
- Identify the success criteria
- List assumptions, risks, obstacles

2. Develop detail Plan
- Identify project activities
- Estimate activity duration
- Determine resource requirements
- Construct / analyze project network
- Prepare project proposal

3. Launch the Plan
- Recruit and organize project team
- Establish team operating rules
- Level resources
- Schedule work packages
- Document work packages

4. Monitor/ Control Progress
- Establish progress reporting system
- Install change control tools/process
- Define problem-escalation process
- Monitor project progress vs plan
- Revise Plan

5. Close Out
- Obtain client acceptance
- Install project deliverables
- Complete project documentation
- Complete post-implementation audit
- Issue final project report

Knowledge Areas for Project Management

1. Project **Integration** Management
2. Project **Scope** management
3. Project **Time** Management
4. Project **Cost** Management
5. Project **Quality** Management
6. Project **Human Resource** Management
7. Project **Communications** Management
8. Project **Risk** Management
9. Project **Procurement** Management

[PMBOK PMI]

1. Project Planning

Project Planning Tools

Critical Path Method, Gantt charts, Activity bar charts

- Build a work-plan from activity data.
- Display work-plan in graphical or tabular form.

Project planning software (e.g., Microsoft Project)

- Maintain a database of activities and related data
- Calculate and display schedules
- Manage progress reports

1. Software Project Planning

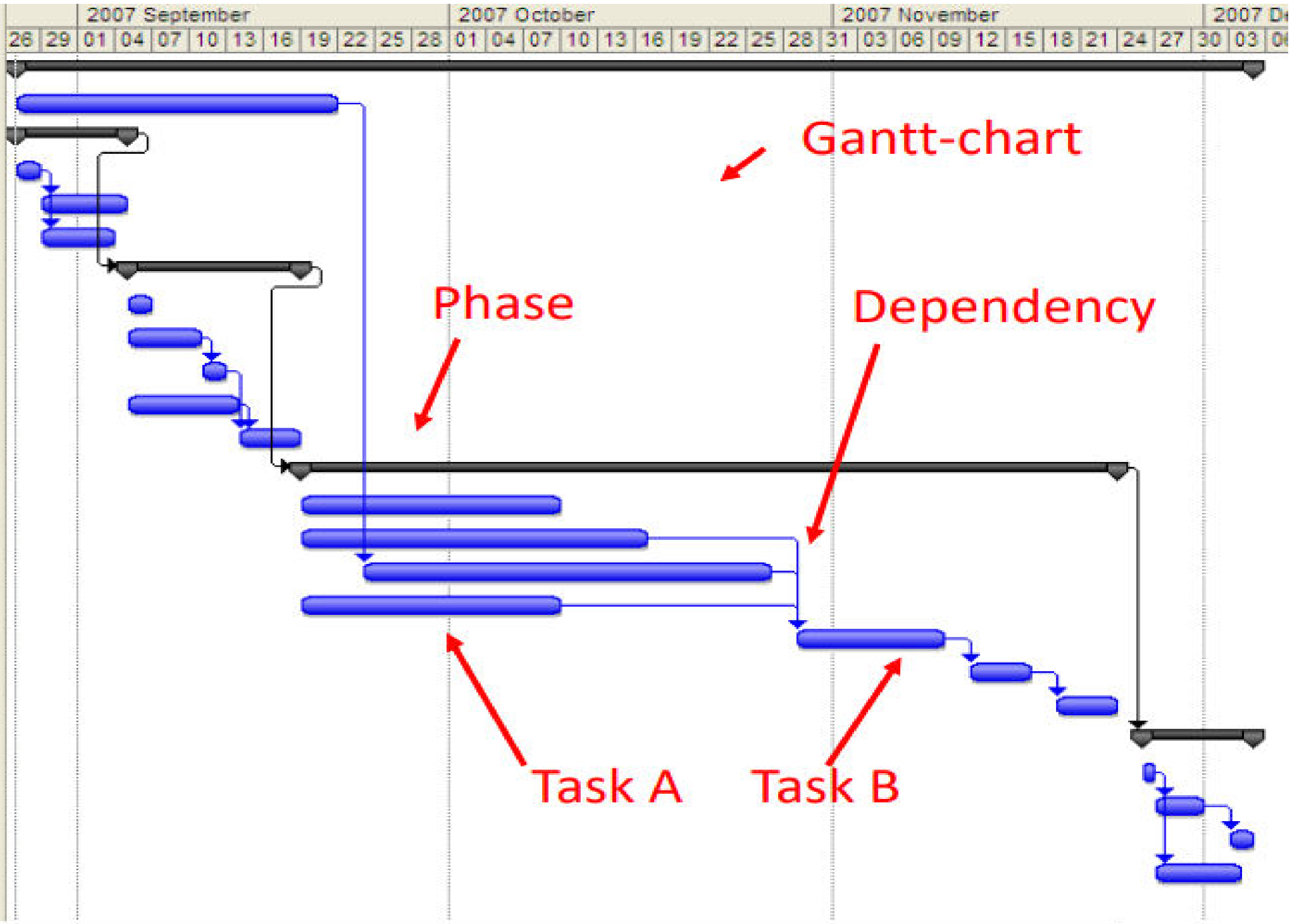
The purpose of the Project Plan is to

- define and establish the management strategy for achieving the goals of the project.

The project development plan is used to:

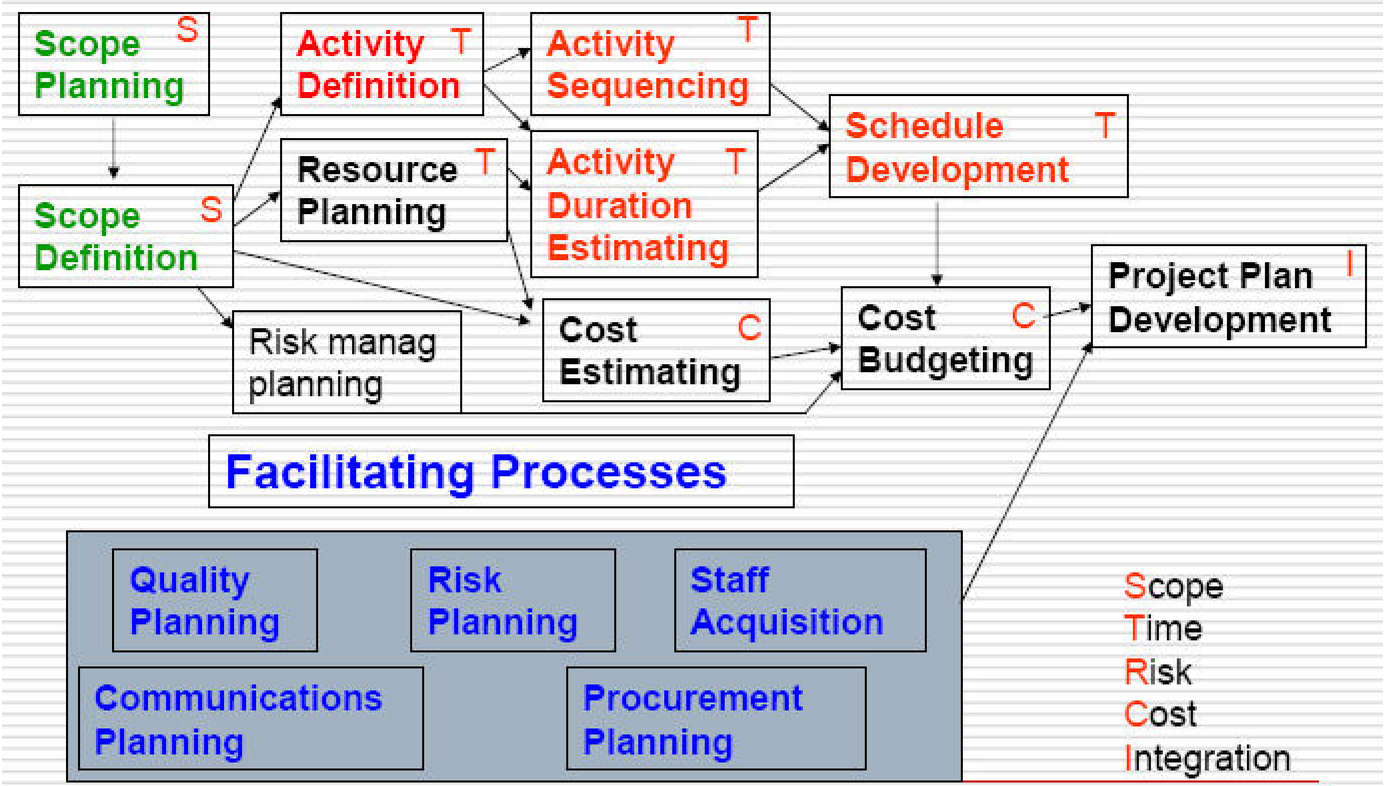
- Guide project execution.
- Document project planning assumptions.
- Document project planning decisions regarding alternatives chosen.
- Facilitate communication among stakeholders.
- Define key management reviews (as to content, extent, and timing).
- Provide a baseline for progress measurement and project control.

Tasks, duration, and dependencies



Task Name	Duration	Start	Finish	Predeces
Project Total	**72 days**	**Mon 07-08-27**	**Tue 07-12-04**	
Training in J2EE	20 days	Mon 07-08-27	Fri 07-09-21	
Inception	**7 days**	**Mon 07-08-27**	**Tue 07-09-04**	
Interview users	2 days	Mon 07-08-27	Tue 07-08-28	
Requirement specification v1	5 days	Wed 07-08-29	Tue 07-09-04	4
User interface prototype	4 days	Wed 07-08-29	Mon 07-09-03	4
Elaboration	**10 days**	**Wed 07-09-05**	**Tue 07-09-18**	3
Finalize project plan	2 days	Wed 07-09-05	Thu 07-09-06	
Requirement specification v2	4 days	Wed 07-09-05	Mon 07-09-10	
Validate Requirements	2 days	Tue 07-09-11	Wed 07-09-12	9
Architecture specification	7 days	Wed 07-09-05	Thu 07-09-13	
Buffer	3 days	Fri 07-09-14	Tue 07-09-18	10;11
Construction	**48 days**	**Wed 07-09-19**	**Fri 07-11-23**	7
Detailed design	15 days	Wed 07-09-19	Tue 07-10-09	
Implement Database module	20 days	Wed 07-09-19	Tue 07-10-16	
Implement Business logic	25 days	Mon 07-09-24	Fri 07-10-26	2
Implement User interface	15 days	Wed 07-09-19	Tue 07-10-09	
Integration testing	10 days	Mon 07-10-29	Fri 07-11-09	15;17;16
System testing	5 days	Mon 07-11-12	Fri 07-11-16	18
Buffer	5 days	Mon 07-11-19	Fri 07-11-23	19
Transition	**7 days**	**Mon 07-11-26**	**Tue 07-12-04**	13
Installation	1 day	Mon 07-11-26	Mon 07-11-26	
Fault correction	4 days	Tue 07-11-27	Fri 07-11-30	22
Acceptance testing	2 days	Mon 07-12-03	Tue 07-12-04	23
User training	5 days	Tue 07-11-27	Mon 07-12-03	22

Planning Process Flow



Types of Plans

Software development plan :The central plan, which describes how the system will be developed.

Quality assurance plan : Specifies the quality procedures & standards to be used.

Validation plan. Defines how a client will validate the system that has been developed.

Types of Plans

Configuration management plan : Defines how the system will be configured and installed.

Maintenance plan : Defines how the system will be maintained.

Staff development plan: Describes how the skills of the participants will be developed.

The Project Plan- Format

Contents

Many format; common sections are: (kathy -62)

Introduction or Overview
- **Project Name, Description and Need it addresses, Sponsor's name, Name of PM, Deliverables (Brief), Reference Material, List of definitions.**

Project Organization
- **Org. chart, Responsibilities, Other Org. or process related info.**

Management and Technical Approach
- **Manage Objectives, project control, Risk Manage, Staffing, Technical Processes.**

Project Scope
- **Major Work Packages/WBS, Key Deliverables, Other Work related Info.,**

Project Schedule
- **Summary/Detail Schedules**

Budget
- **Summary/Detail budget,**

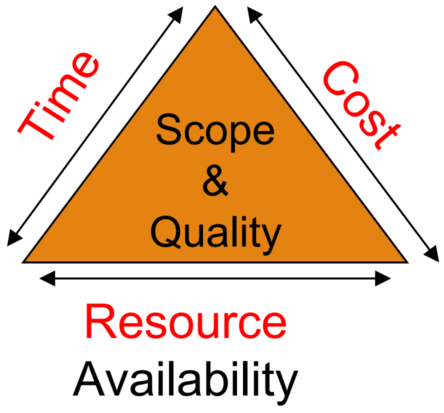
2. Scope Definition

The first step - define the Scope

"Scope" **The term may refer to:**

a) **Product scope** **- the features and functions that are to be included in a product or service.**

b) **Project scope** **- the work that must be done in order to deliver a product (with the specified features and functions).**



1. ***Product Scope*** **is defined in the product requirements and is the subject of the product life cycle**
2. ***Project Scope*** **is defined in the project charter and is the subject of the project plan**

(Kathy-ch4)

Decomposition

Subdividing the major project deliverables into smaller, more manageable Components (products) in sufficient detail

- **to support future project activities (planning, executing, controlling, and closing)**

Requires Steps:

1) **Identify the major elements (deliverable) of the project.**
2) **Identify constituent elements (Work Products) of the deliverable.**
3) **Decide if adequate cost and duration estimates can be developed at this level of detail for each element.**

WBS-Work Breakdown Structure

- Overall work has to be decomposed into manageable units
- Complex tasks are broken down into subtasks and further refined called Work Breakdown Structures (WBS)

WBS-Work Breakdown Structure

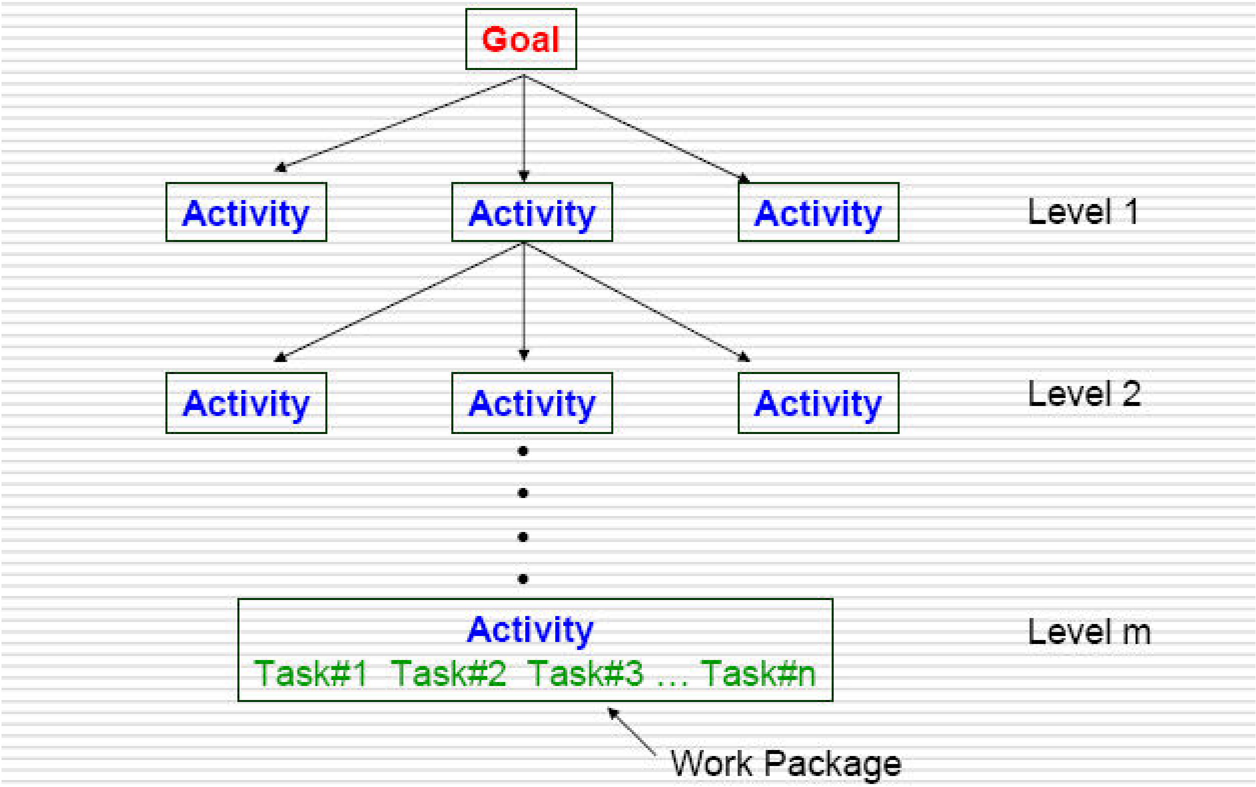
- There are many ways of breaking down the activities in a project, but the most usual is into:
  - work packages;
  - tasks/Activities
  - deliverables/Work Products
  - milestones

WBS Framework

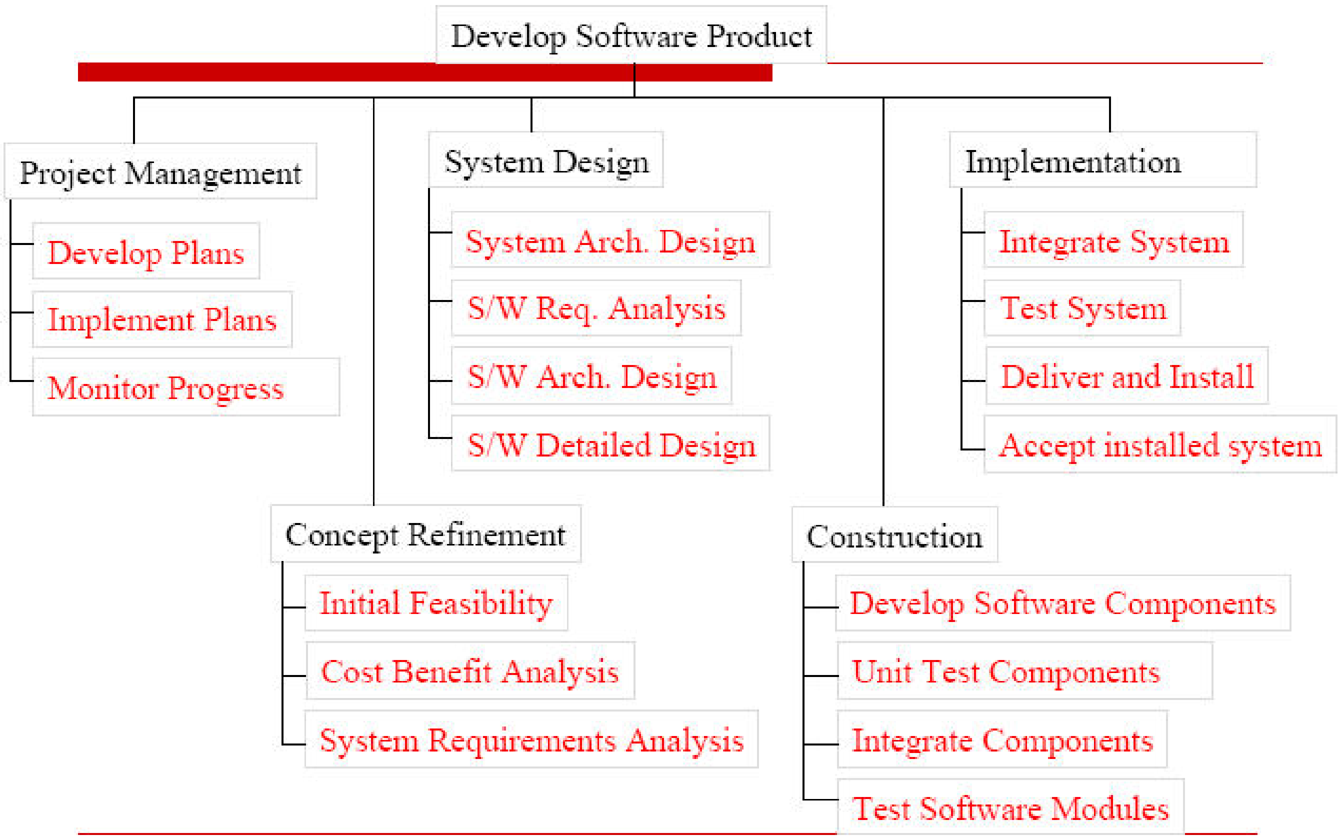
A framework dictating the number of levels and the nature of each level may be imposed on a WBS (e.g. IBM recommended **five levels):**

- **Level 1:** Project.
- **Level 2:** Deliverables (such as software, manuals and training courses)
- **Level 3:** Components; Which are the key work items needed to produce deliverables (e.g. modules and tests required to produce the system software)
- **Level 4:** Activities (**Work-packages** which are major work items, or collections of related tasks, required to produce a component)
- **Level 5:** Tasks (tasks that will normally be the responsibility of a single person).

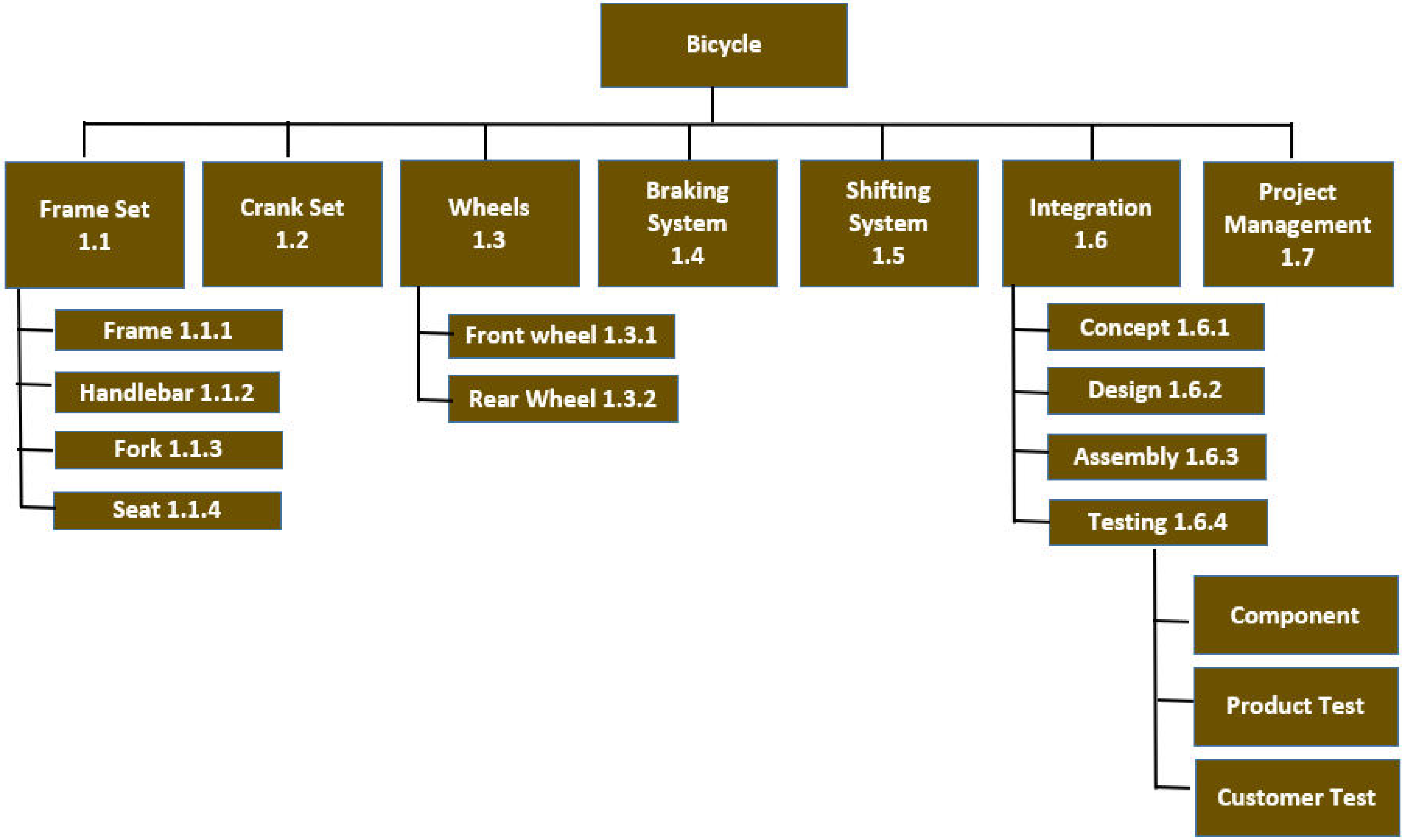
Hierarchy of WBS



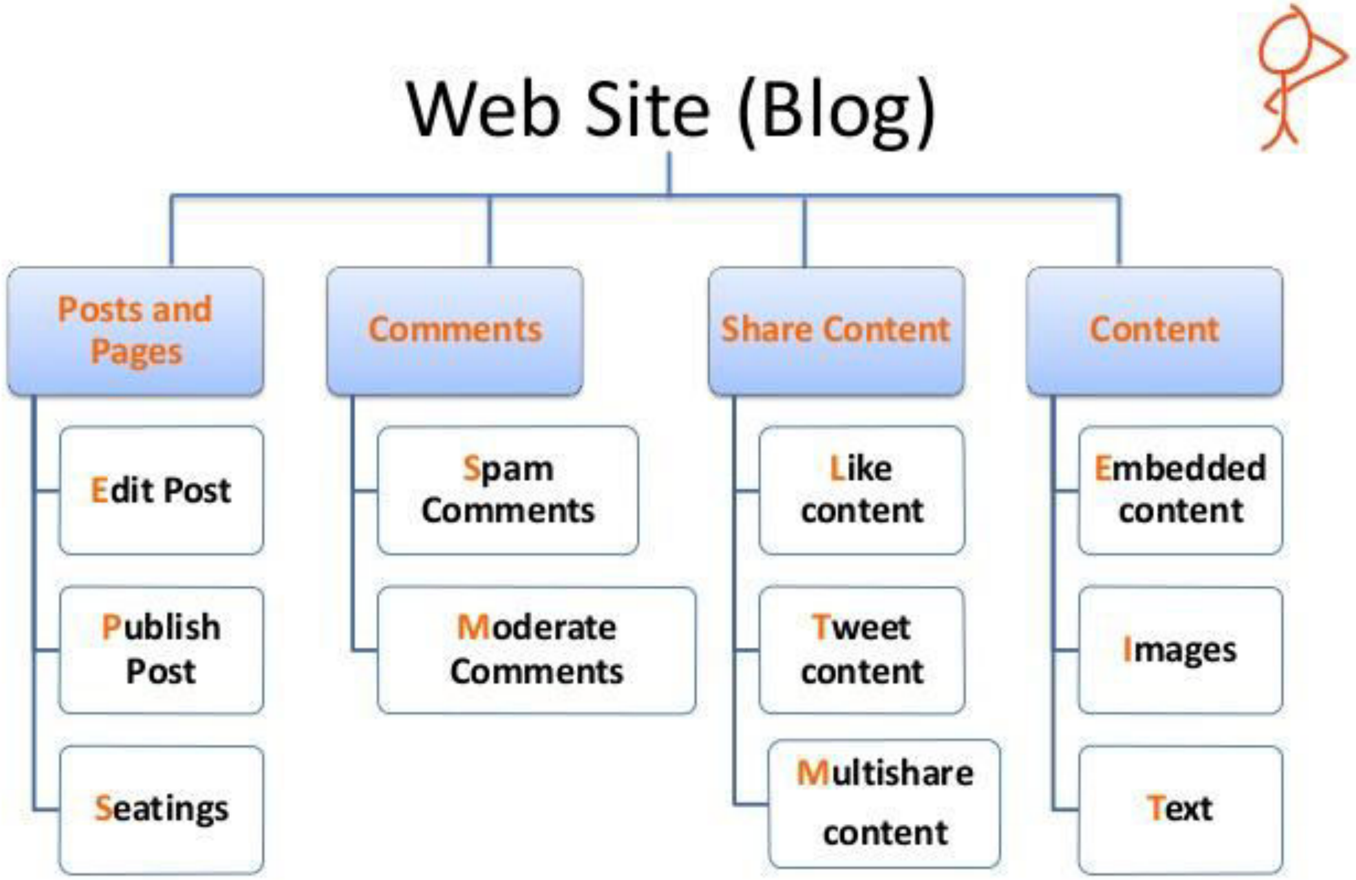
WBS Template for Software Development Project



Example


Bicycle
Frame Set 1.1
Crank Set 1.2
Wheels 1.3
Braking System 1.4
Shifting System 1.5
Integration 1.6
Project Management 1.7
Frame 1.1.1
Handlebar 1.1.2
Fork 1.1.3
Seat 1.1.4
Front wheel 1.3.1
Rear Wheel 1.3.2
Concept 1.6.1
Design 1.6.2
Assembly 1.6.3
Testing 1.6.4
Component
Product Test
Customer Test

Examples



Web Site (Blog)
Posts and Pages
Comments
Share Content
Content
Edit Post
Publish Post
Seatings
Spam Comments
Moderate Comments
Like content
Tweet content
Multishare content
Embedded content
Images
Text

3. Activity Planning

Planning And Scheduling

- Need to develop a schedule in which planned start and end dates are assigned to all activities
  - Activity precedence & Dependencies
  - Activity duration
  - Available resources

Steps

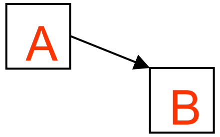
Define activities (Gantt chart)

Sequence activities (CPM)

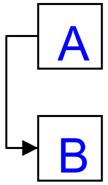
Estimate time (PERT)

Task Dependencies

Task Dependencies (relationships)

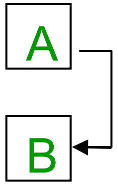


Finish-to-start (FS): Task (B) cannot start until task (A) finishes

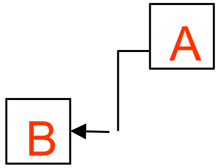


Start-to-start (SS): Task (B) cannot start until task (A) starts

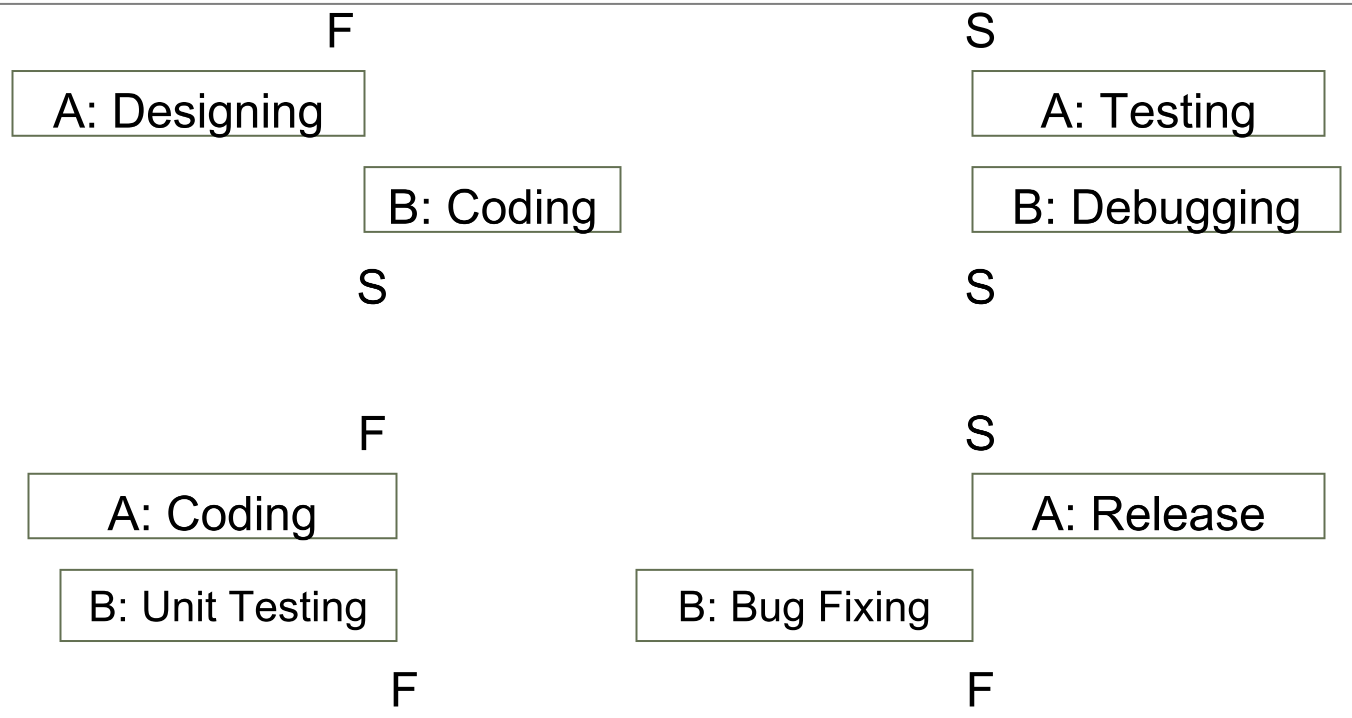
Finish-to-Finish (FF): Task (B) cannot finish until task (A) finishes



Start-to-finish (SF): Task (B) cannot finish until task (A) starts



Scheduling as precedence

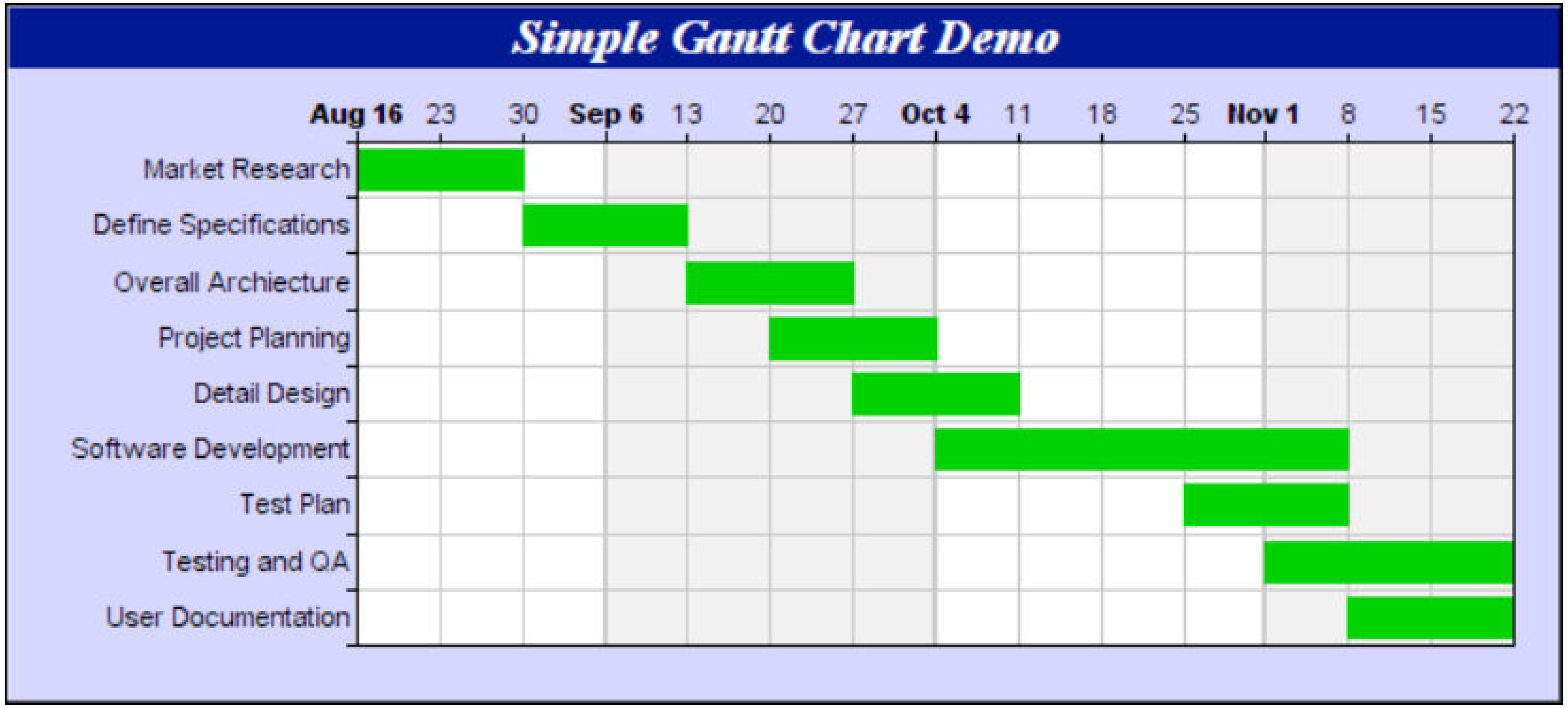


Gantt chart - Define activities

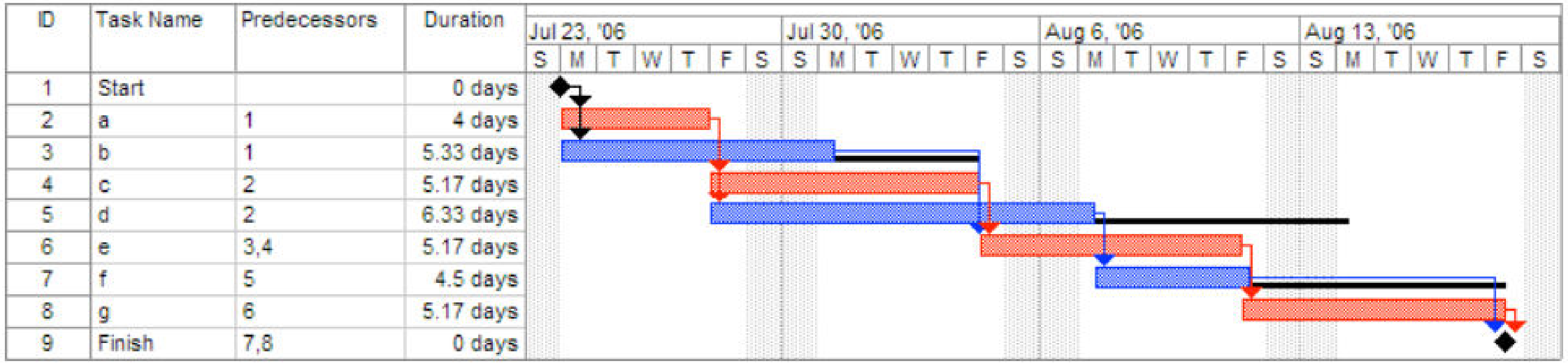
Developed in 1918 by H.L. Gantt

■ Graph or bar chart with a bar for each project activity that shows passage of time ■ Provides visual display of project schedule

Gantt Chart


Simple Gantt Chart Demo
Aug 16
23
30
Sep 6
13
20
27
Oct 4
11
18
25
Nov 1
8
15
22
Market Research
Define Specifications
Overall Archiecture
Project Planning
Detail Design
Software Development
Test Plan
Testing and QA
User Documentation

Gantt Chart



ID	Task Name	Predecessors	Duration
1	Start		0 days
2	a	1	4 days
3	b	1	5.33 days
4	c	2	5.17 days
5	d	2	6.33 days
6	e	3,4	5.17 days
7	f	5	4.5 days
8	g	6	5.17 days
9	Finish	7,8	0 days

A Gantt chart created using Microsoft Project. The critical path is in red, and the slack is the black lines connected to non-critical activities.

Since Saturday and Sunday are not work days, some bars on the Gantt chart are longer if they cut through a weekend.

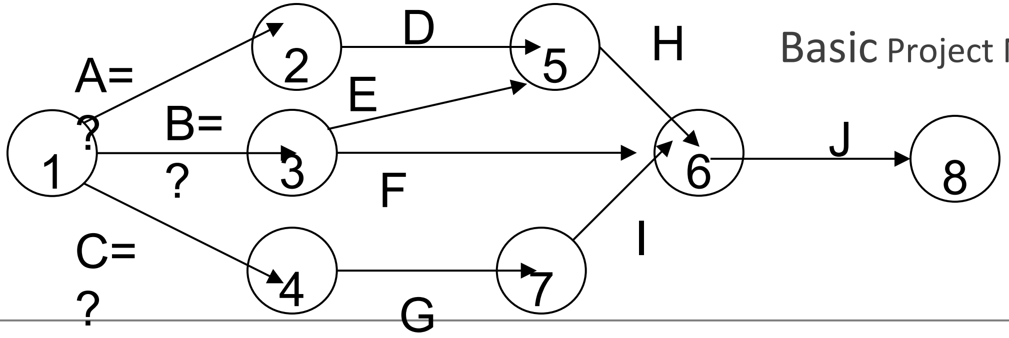
Project Network Diagrams

- ❖ Project network diagrams (PND) are the preferred technique for showing activity sequencing
- ❖ A project network diagram is a schematic display of the logical relationships among, or sequencing of, project activities

- **Arrow Diagramming Method (ADM)**
  - Also called activity-on-arrow (AOA) project network diagrams
  - Activities are represented by arrows
  - Nodes or circles are the starting and ending points of activities
  - Can only show finish-to-start dependencies

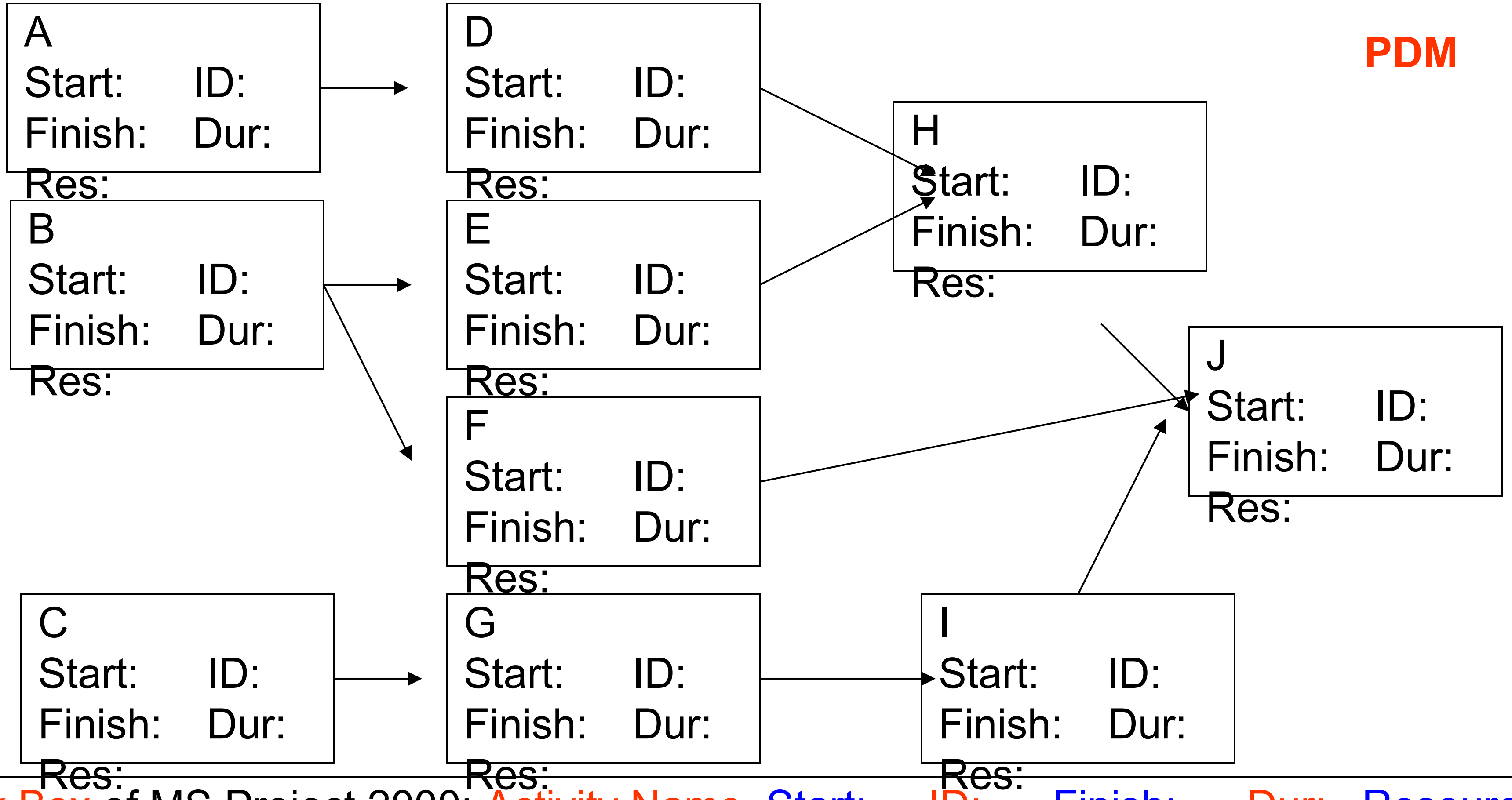
- **Precedence Diagramming Method (PDM)**
  - Activities are represented by boxes
  - Arrows show relationships between activities
  - More popular than ADM method and used by project management software
  - Better at showing different types of dependencies

Basic Project Network Diagrams (PND).....



AOA (ADM);
CPM adds more inf at nodes

PDM



Task Box of MS Project 2000: Activity Name Start: ID: Finish: Dur: Resource:

Terminology: Network Diagram

Path

- Sequence of activities that leads from the starting node to the finishing node

Critical path

- The longest path; determines expected project duration

Critical activities

- Activities on the critical path

Slack

- Allowable slippage for path; the difference the length of path and the length of critical path

Terminology: Network Diagram

Slack time (also known as float)

- is the amount of delay expressed in units of time that could be tolerated in the starting time or completion time of an activity without causing a delay in the completion of the project.

Slack time is the difference between the **late finish** and the **early finish** (LF-EF).

If the result is greater than zero, then the activity has a range of time in which it can start and finish without delaying the project completion date

Sequence activities (Critical Path Method CPM)

project network analysis technique used to predict total project Duration

primary objectives:

- Planning the project to complete it as quickly as possible
- Identifying those activities where a delay in their execution
  - affect the overall end date of the project or ‘later activities’ start dates.

- represents activities as links (arrowed lines) in the graph
- nodes (circles) represent the events of activities start and finish.
- A link represents an activity (and has duration)

- A slack period is **one in which there is not much work or activity**.
  - **Slack time = Latest start time - Earliest start time**.

Critical path method

A *forward pass* analysis performs schedule calculations that identify the early start and finish dates of activities and the project

A *backward pass* analysis performs schedule calculations that identify the late start and finish dates of activities and the project, as well as total and free float

Total float (TF) is the amount of time an activity can be delayed without delaying the project as a whole

Free float (FF) is the amount of time an activity can be delayed without delaying its successor (dependent) activities

Note that total float is global to the project, while free float is local to the neighborhood of the activity

Terminology: CPM

Network activities

- ES: early start
- EF: early finish
- LS: late start
- LF: late finish

Used to determine

- Expected project duration
- Slack time
- Critical path

CPM –Forward Pass

ES : 1

EF : ES +Duration - 1

ES of Next Activity : EF of previous activity +1

EF of next Activity : ES+Duration-1

CPM – Backsword Pass

LF = EF of last activity

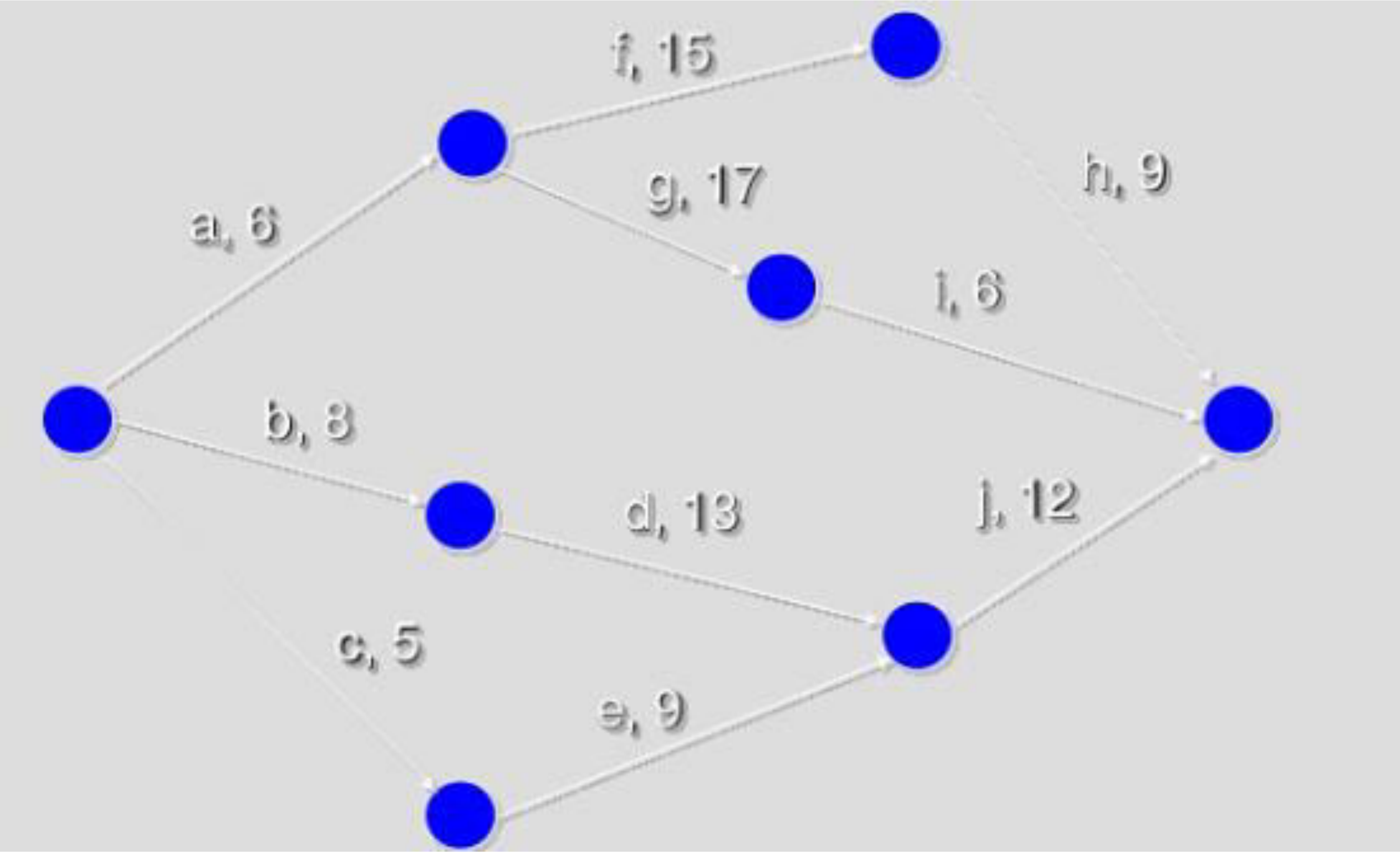
LS = LF-Duration+1

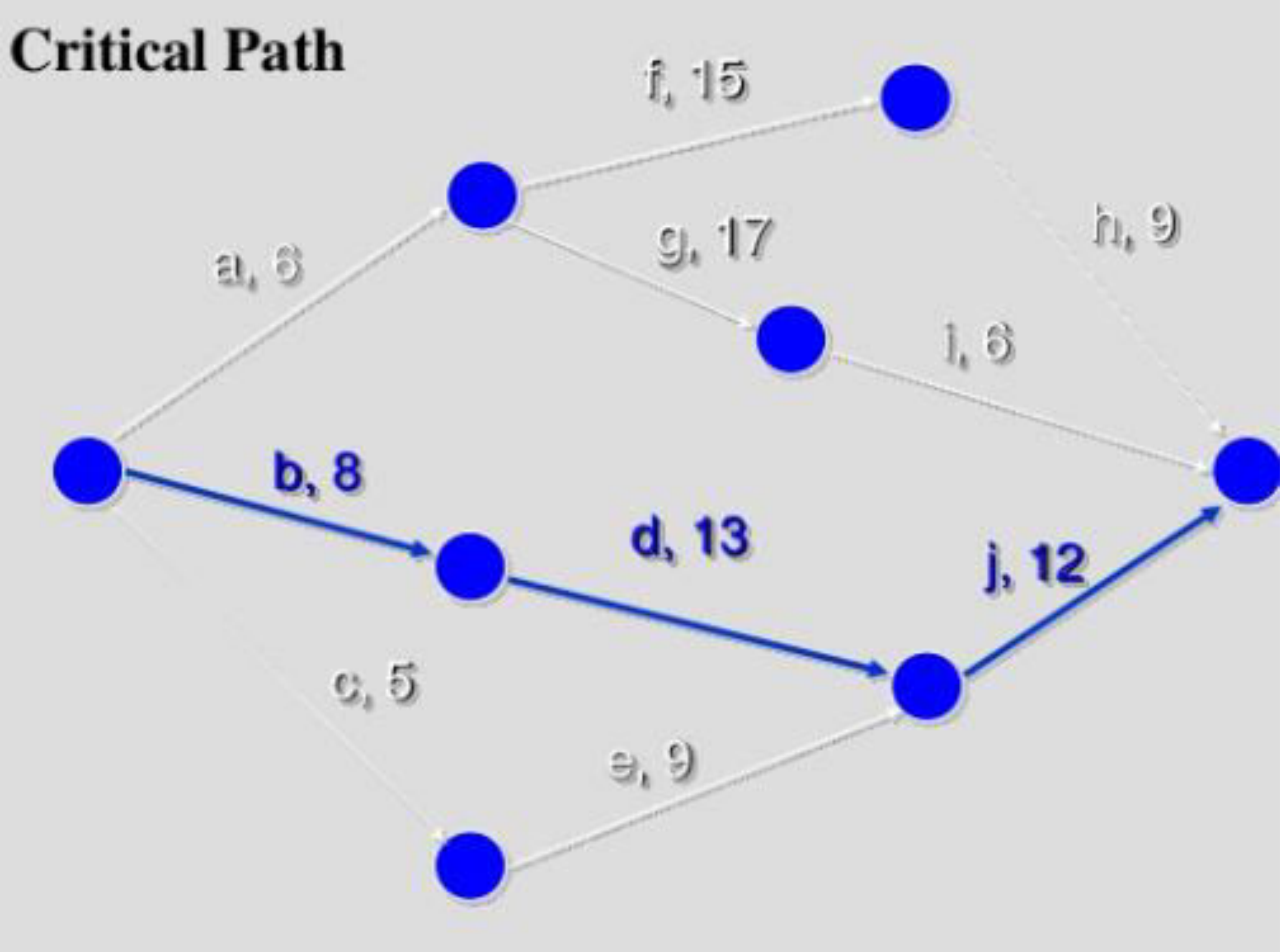
LF of prev Activity = LS -1 of Prev Activity

https://www.pmexamsmartnotes.com/how-to-calculate-critical-path-float-and-early-and-late-starts-and-finishes/3/

Example

Activity	Duration	Precedence
A	6	
B	8	
C	5	
D	13	B
E	9	C
F	15	A
G	17	A
H	9	F
I	6	G
J	12	D,E


f, 15
h, 9
g, 17
a, 6
i, 6
b, 8
d, 13
j, 12
c, 5
e, 9


Critical Path
f, 15
a, 6
g, 17
h, 9
i, 6
b, 8
d, 13
j, 12
c, 5
e, 9


Thank You

Activity		Duration (weeks)	Precedents
A	Hardware selection	6	
B	Software design	4	
C	Install hardware	3	A
D	Code & test software	4	B
E	File take-on	3	B
F	Write user manuals	10	
G	User training	3	E
H	Install & test system	2	C